

Report: Security Hardening

1. Introduction

Security hardening in Active Directory is crucial to reduce attack surfaces and enforce stricter authentication mechanisms. In this task, key configurations such as enabling LDAP signing, disabling NTLM authentication, and implementing LAPS (Local Administrator Password Solution) were carried out to increase the domain's resistance to common exploitation techniques. These changes help ensure the confidentiality and integrity of communications, while also automating and protecting local administrator credentials across all domain-joined machines

2. Objective

The objective of this task was to enhance the domain's overall security posture by applying foundational hardening measures that protect against unauthorized access and credential abuse. This included enforcing LDAP signing to ensure secure communications between clients and domain controllers, disabling outdated and insecure NTLM authentication protocols, and deploying Microsoft LAPS to automatically manage and securely store local administrator passwords. These configurations were intended to reduce common vulnerabilities and align the environment with modern security best practices.

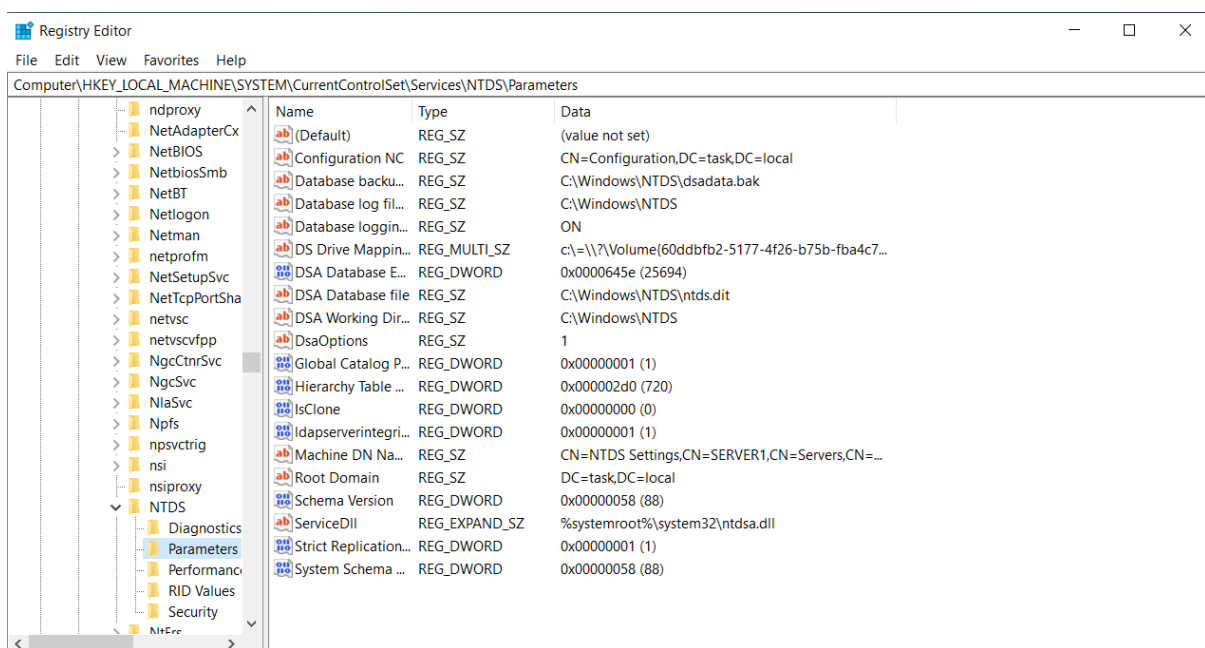
3. Tools used

- Group Policy Management Console (GPMC)
- Registry Editor
- Windows PowerShell

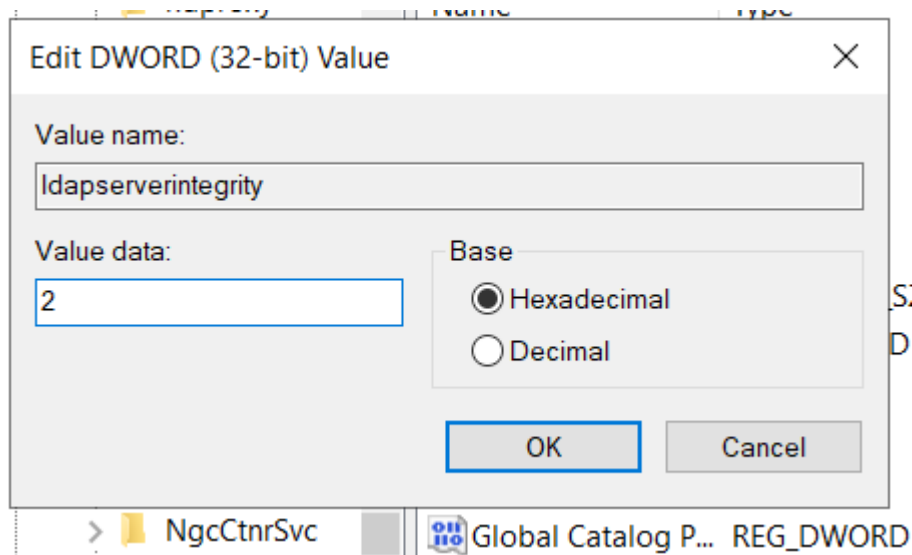
4. Methodology

4.1 Enable LDAP Signing

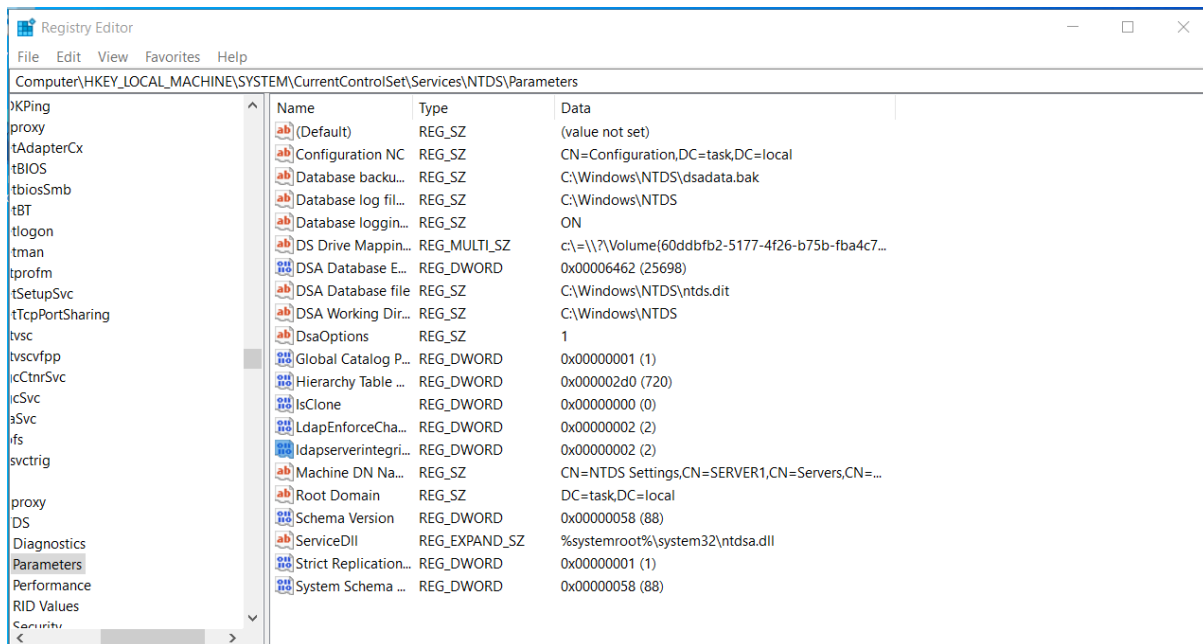
LDAP signing was configured by modifying the following registry key: HKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\Services\NTDS\Parameters. The value of ldapserverintegrity was changed from 1 to 2 to enforce that LDAP signing is required. Additionally, a new DWORD value named LdapEnforceChannelBinding was created and set to 2 to enable channel binding enforcement.



(Navigate to NDTs parameters in Registry Editor)



(The value LDAPServerIntegrity is already there so Change its value from 1 to 2)



(And add one DWORD LdapEnforceChannelBinding)

4.2 Disable NTLM Authentication

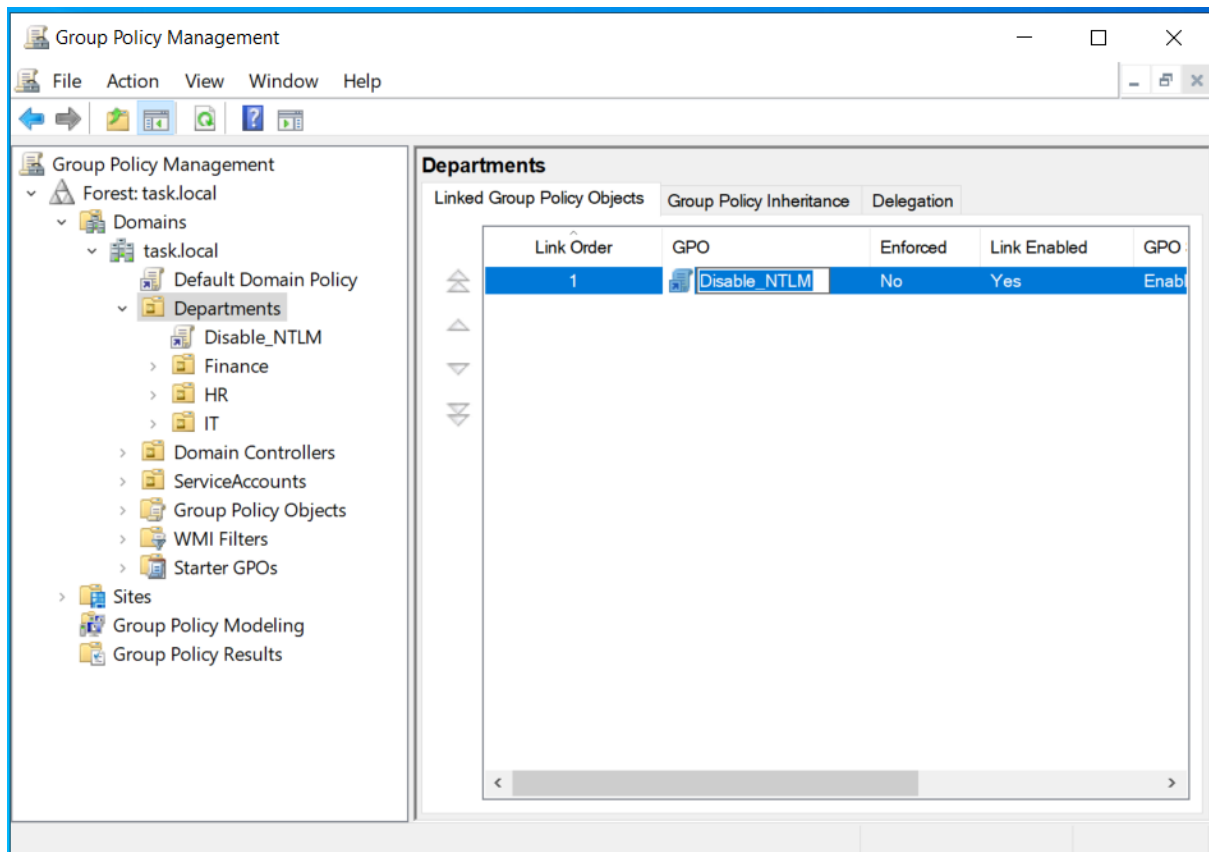
A new Group Policy Object named Disable_NTLM was created and edited under:

Computer Configuration → Windows Settings → Security Settings → Local Policies → Security Options

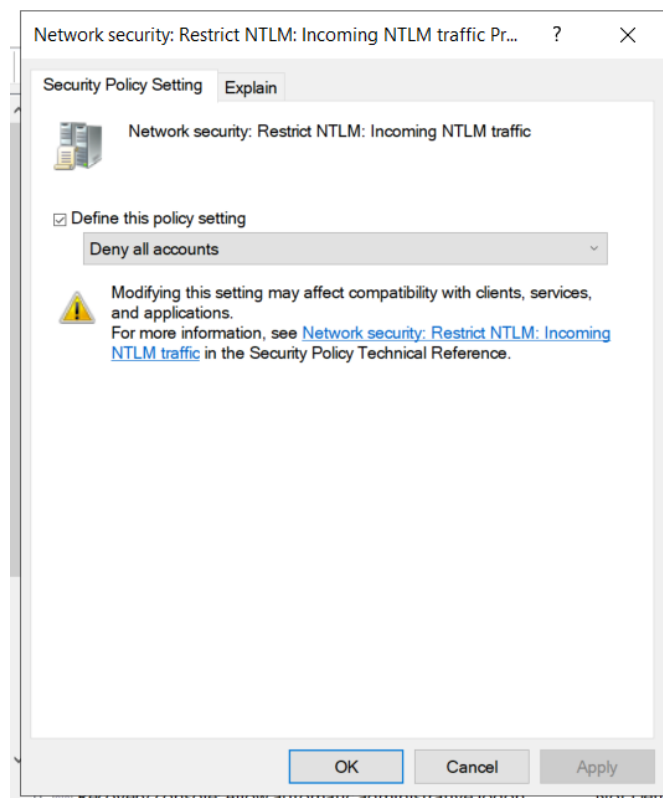
The following settings were configured:

- Network security: Restrict NTLM: Incoming NTLM traffic → Deny all accounts
- Network security: Restrict NTLM: Outgoing NTLM traffic to remote servers → Deny all

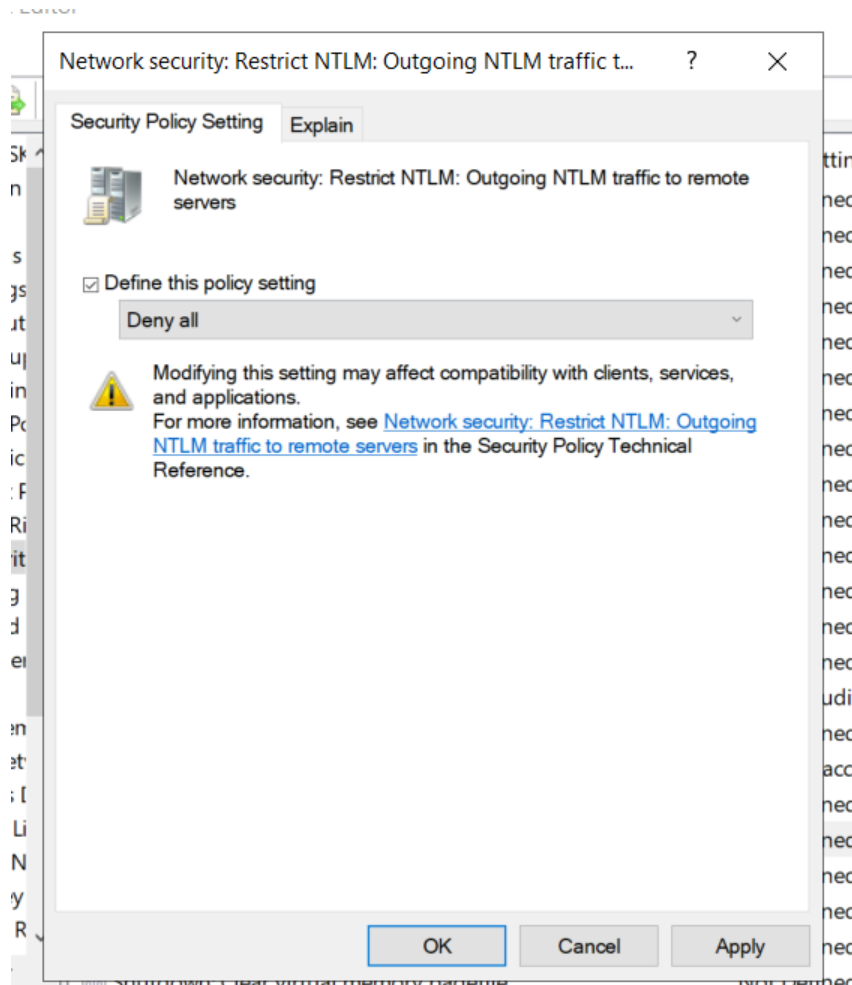
This GPO was then linked to the Domain Controllers OU to apply restrictions specifically on domain controller authentication paths.



(A new GPO is created and linked with departments)



(Edit policy)



(Edit policy)

```
Microsoft Windows [Version 10.0.19044.5737]
(c) Microsoft Corporation. All rights reserved.

C:\Users\administrator>gpupdate /force
Updating policy...

Computer Policy update has completed successfully.
User Policy update has completed successfully.

C:\Users\administrator>
```

(Policy is updated on PC1)

```

C:\Users\administrator>gpresult /r

Microsoft (R) Windows (R) Operating System Group Policy Result tool v2.0
© Microsoft Corporation. All rights reserved.

Created on 5/13/2025 at 1:29:39 PM

RSOP data for TASK\Administrator on PC1 : Logging Mode
-----

OS Configuration:          Member Workstation
OS Version:                10.0.19044
Site Name:                 Default-First-Site-Name
Roaming Profile:           N/A
Local Profile:             C:\Users\administrator
Connected over a slow link?: No

COMPUTER SETTINGS
-----
CN=PC1,OU=Finance,OU=Departments,DC=task,DC=local
Last time Group Policy was applied: 5/13/2025 at 1:28:10 PM
Group Policy was applied from:   server1.task.local
Group Policy slow link threshold: 500 kbps
Domain Name:                   TASK
Domain Type:                   Windows 2008 or later

Applied Group Policy Objects
-----
  Disable_USB_Finance
  Logon_Audit_Policy
  Object_Access_Audit
  ScreenLock_15Min
  Disable_SMBv1
  Logon_Banner_GPO
  Disable_NTLM
  Default Domain Policy

```

(Disable NTLM is applied successfully)

4.3 Implement LAPS

A cumulative security update was installed to enable native Windows LAPS functionality. The Laps PowerShell module was imported, and the schema was extended using Update-LapsADSchema. All necessary attributes, including ms-LAPS-Password, ms-LAPS-PasswordExpirationTime, and others, were successfully added. A new GPO was created

and configured under Computer Configuration → Administrative Templates → System → LAPS with the following settings:

- Enable password backup for DSRM accounts
- Enable password encryption
- Configure authorized password decryptors
- Name of administrator account to manage
- Configure password backup directory
- Password Settings

This GPO was linked to the Departments OU, and Group Policy was applied using gpupdate /force on the client.

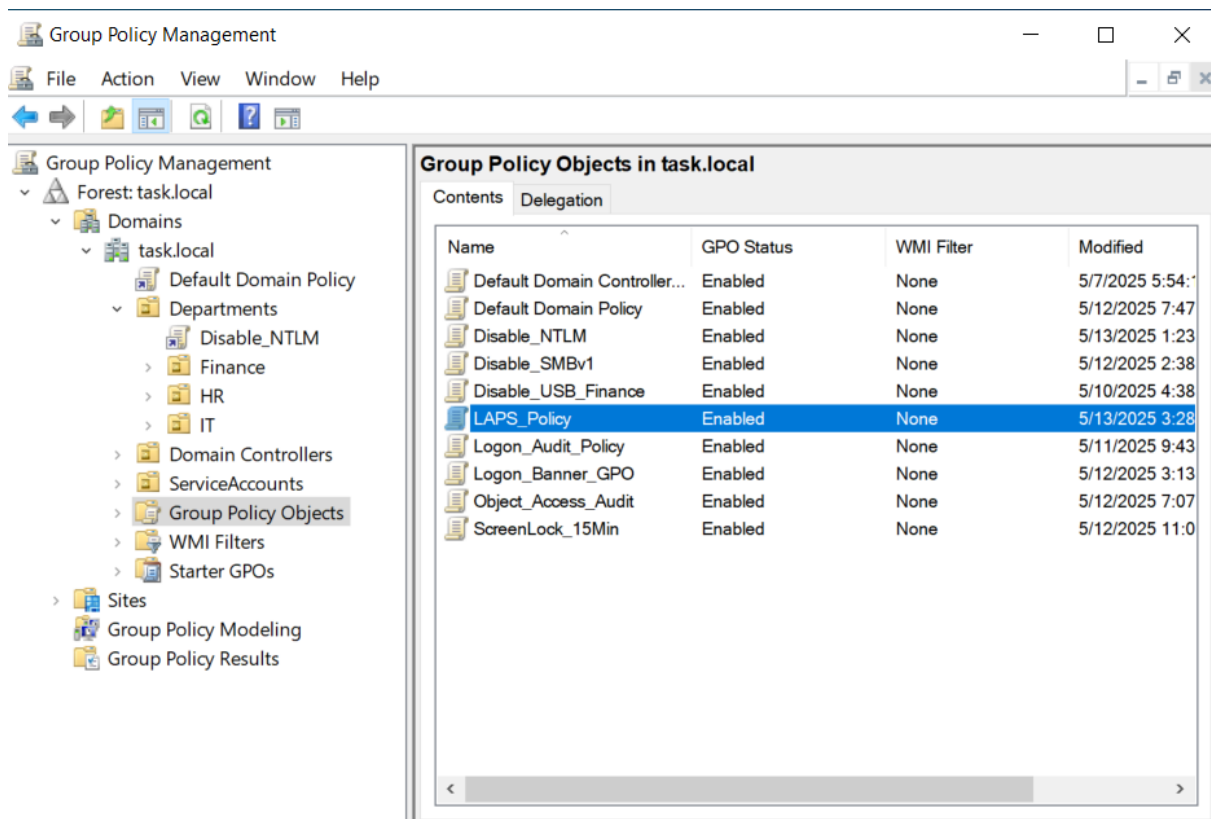
 Administrator: Windows PowerShell

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\Administrator> Import-Module Laps
PS C:\Users\Administrator> Update-LapsADSchema
```

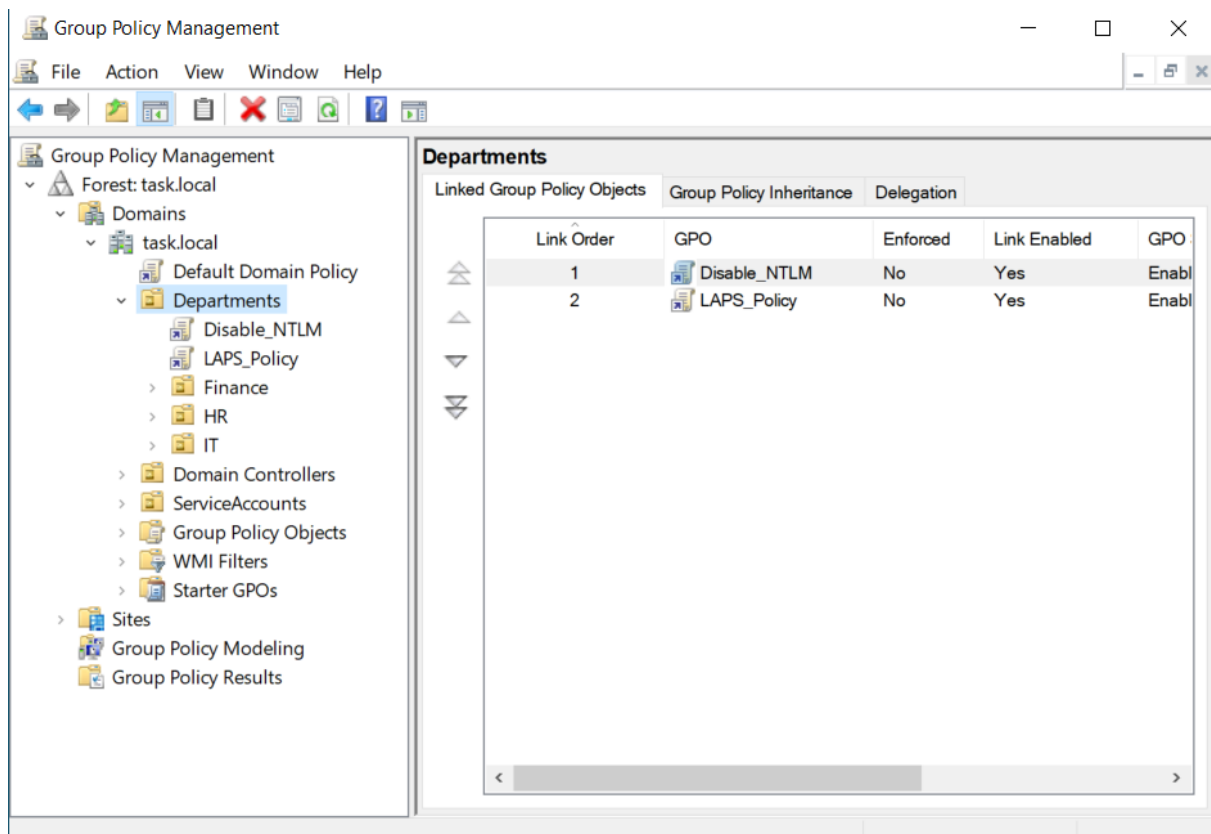
(Imported and updated LAPS module)



(Created a new GPO – LAPS_Policy)

Setting	State	Comment
Enable password backup for DSRM accounts	Enabled	No
Configure size of encrypted password history	Not configured	No
Enable password encryption	Enabled	No
Configure authorized password decryptors	Enabled	No
Name of administrator account to manage	Enabled	No
Configure password backup directory	Enabled	No
Do not allow password expiration time longer than required ...	Not configured	No
Password Settings	Enabled	No
Post-authentication actions	Not configured	No

(Edit policy)



(Link GPO to departments)

C:\> Administrator: Command Prompt

Applied Group Policy Objects

```
-----  
Disable_USB_Finance  
Logon_Audit_Policy  
Object_Access_Audit  
ScreenLock_15Min  
Disable_SMBv1  
Logon_Banner_GPO  
Disable_NTLM  
LAPS_Policy  
Default Domain Policy
```

The following GPOs were not applied because they were filtered out

```
-----  
Local Group Policy  
Filtering: Not Applied (Empty)
```

The computer is a part of the following security groups

```
-----  
BUILTIN\Administrators  
Everyone  
BUILTIN\Users  
NT AUTHORITY\NETWORK  
NT AUTHORITY\Authenticated Users
```

(LAPS is applied successfully)

Administrator: Windows PowerShell

```
PS C:\Users\Administrator> Get-LapsADPassword -Identity PC1
```

```
ComputerName      : PC1  
DistinguishedName : CN=PC1,OU=Finance,OU=Departments,DC=task,DC=local  
Account           : Administrator  
Password           : System.Security.SecureString  
PasswordUpdateTime : 5/13/2025 5:14:06 PM  
ExpirationTimestamp : 6/12/2025 5:14:06 PM  
Source             : EncryptedPassword  
DecryptionStatus   : Success  
AuthorizedDecryptor : TASK\Administrator
```

```
PS C:\Users\Administrator>
```

(The Get-LapsADPassword -Identity PC1 command works, means LAPS setup is fully functional)

5. Recommendations

- Regularly audit the LAPS-related attributes in Active Directory to ensure password storage and expiration are functioning as intended.
- Limit access to the Get-LapsADPassword cmdlet by assigning permissions only to authorized administrative users or groups.
- Monitor LDAP and NTLM-related logs to identify any fallback attempts or misconfigurations across the network.
- Keep Group Policy Objects regularly reviewed and version-controlled to detect any unauthorized changes or rollbacks.
- Enable alerting mechanisms through SIEM tools to track LAPS password access and schema changes in real time.
- Maintain patch compliance on all domain controllers to ensure support for the latest LAPS features and hardening settings.

6. Conclusion

The security hardening configuration was successfully completed by enforcing LDAP signing, disabling legacy NTLM protocols, and implementing Microsoft LAPS. These measures collectively reduce attack exposure, increase password security, and ensure that credentials are no longer stored or transmitted insecurely. Each setting was verified either via Registry, GPO status, or PowerShell output, confirming correct implementation without any policy conflicts or errors.